



Anterior segment optical coherence tomography (AS-OCT) image analysis methods and applications: A systematic review

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ABSTRACT

Background: Anterior segment optical coherence tomography (AS-OCT) constitutes an important imaging modality to examine the anterior eye, which is commonly used in research and clinical practice. Since its introduction, a range of image analysis methods have been developed to quantify these images using different analysis techniques for various applications. This systematic review aims to provide an in-depth summary and to classify image analysis techniques found in the literature applied to AS-OCT images.

Methods: Scopus and Engineering Village databases were searched to retrieve relevant studies up to and including January 2022. Customized search statements were used along with cross reference and hand search techniques to ensure a complete coverage. Performance metrics were extracted, analyzed, and compared (when possible).

Results: Three main application categories were identified: glaucoma assessment, corneal segmentation, and anterior segment biometry. These three categories constitute 66% of the total studies reported in this review. Studies were also analyzed by year of publication, and since 2019 deep learning methods were favored over traditional programming or machine learning methodologies. Overall, the AS-OCT image analysis field is less developed compared to posterior segment OCT imaging.

Conclusion: This review presents the state of the art in the field of AS-OCT image analysis. It highlights the opportunities for future areas of research, such as the expansion of DL methods and the extension to specific clinical areas that have received limited attention including surgical monitoring, contact lenses, and specific clinical conditions such as keratoconus and corneal lesions.

1. Introduction

Optical coherence tomography (OCT) is a non-invasive imaging technique capable of visualizing biological samples with micron level resolution [1]. From its first in vivo application, using a time-domain configuration [2], the technology has progressively evolved towards the Fourier domain (FD) [3], which encompasses both swept-source and spectral-domain configurations and allows for faster scanning times, higher axial resolution [4], and an enhanced signal-to-noise ratio [5]. Consequently, FD-OCT configurations have become a standard clinical and research tool in the diagnosis and monitoring of ocular disease [6].

OCT imaging of the anterior segment of the eye (AS-OCT), was introduced in 1994 [7], and has continued to advance with respect to resolution, tissue penetration, and analysis techniques [6]. AS-OCT images provide fundamental information for clinicians and researchers to understand the anatomical features of the anterior segment and

related disorders [8]. This imaging technique also facilitates the timely diagnosis of corneal and ocular surface pathologies (e.g. keratoconus, anterior chamber angle closure, ocular surface lesions, and iris tumors [9]), and the ability to monitor the effect of specific treatments [8,10]. Although some clinical information can be extracted from the simple visual inspection of OCT images (i.e. gross morphological changes), the precise objective quantification of ocular structures is desirable to detect subtle anatomical variations over time or in comparison to normative data. Since manual processes used to extract information from AS-OCT images are time-consuming, and liable to suffer from subjective judgement errors [11,12], a range of automatic image analysis techniques have been developed.

Due to their objectivity, repeatability, and accuracy, automated image analysis methods have shown excellent potential for diagnosis (i.e. classification) and segmentation tasks [13,14], and have been applied across a range of medical disciplines [15,16]. Image analysis methods

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